



SOMAIYA
VIDYAVIHAR UNIVERSITY

K J Somaiya School of Engineering
(formerly K J Somaiya College of Engineering)



KJSSE/RAI/TYBTECH /SEM-V/DS/2026-27

Experiment No. : 6

Title: Analyzing and Interpreting Statistical Distributions in Robotic Environments



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Batch: A2
Experiment No.: 6

Roll No.:16014324028

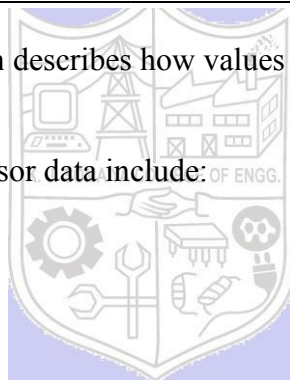
Aim: To analyze and interpret the probability distributions of robotic sensor measurements using graphical and statistical techniques.

Resources needed: Computer/Laptop, Python 3.x, Jupyter Notebook / Google Colab NumPy, Pandas, Matplotlib, SciPy, Robotics sensor dataset

Theory :- A **probability distribution** describes how values of a random variable are distributed.

Common distributions relevant to sensor data include:

- Normal distribution
- Uniform distribution
- Exponential distribution
- Poisson distribution



Sensor data may also exhibit:

- Skewness
- Kurtosis
- Outliers
- Multimodal behavior

Understanding distributions helps determine the characteristics of sensor measurements and can support anomaly detection and statistical modelling.

Activity:

1. Load a robotic sensor dataset.
2. Select suitable sensor attributes.
3. Calculate mean, median and standard deviation.
4. Plot histograms.
5. Generate density plots.
6. Calculate skewness and kurtosis.
7. Identify potential outliers.
8. Compare the observed data with suitable theoretical distributions.
9. Interpret the distribution of each selected sensor variable.
10. Discuss how distribution characteristics may affect robotic data analysis.

Results: (program with output)

The statistical distributions of selected robotic sensor variables were analyzed using numerical and graphical techniques. The shape, spread, skewness and possible outliers were identified.

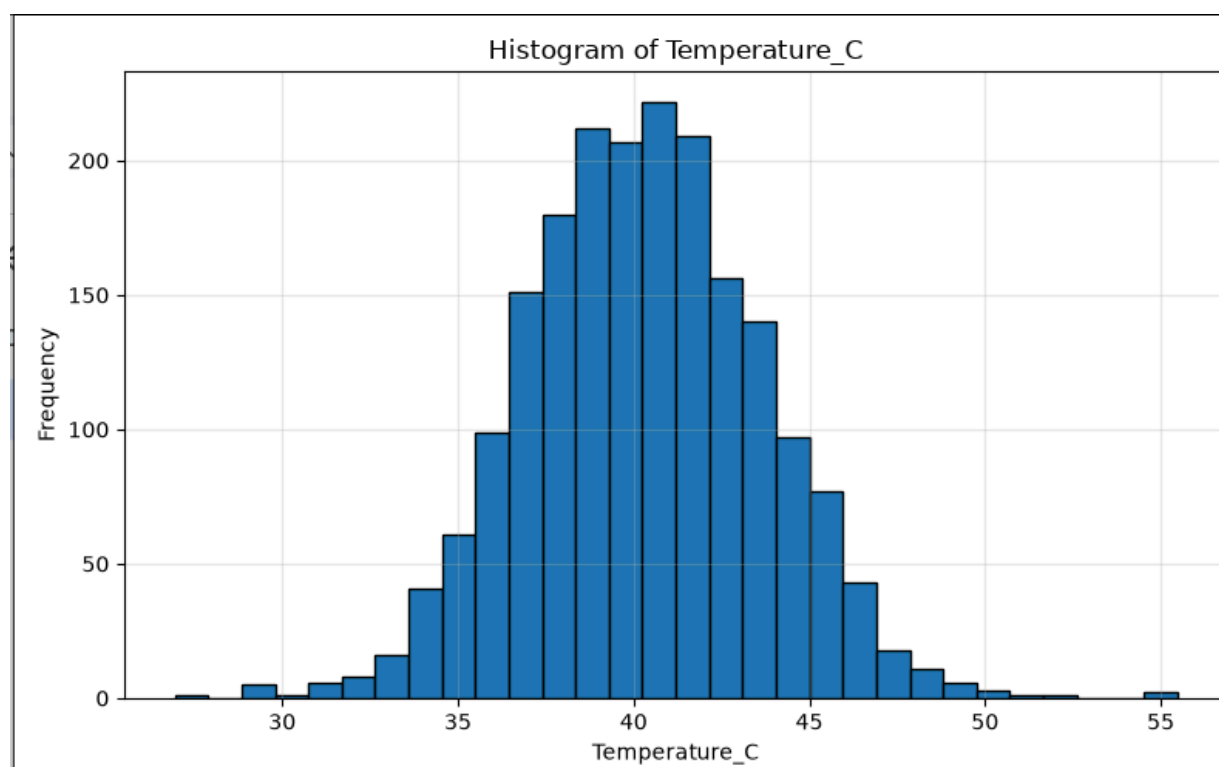
Post-Lab Questions:

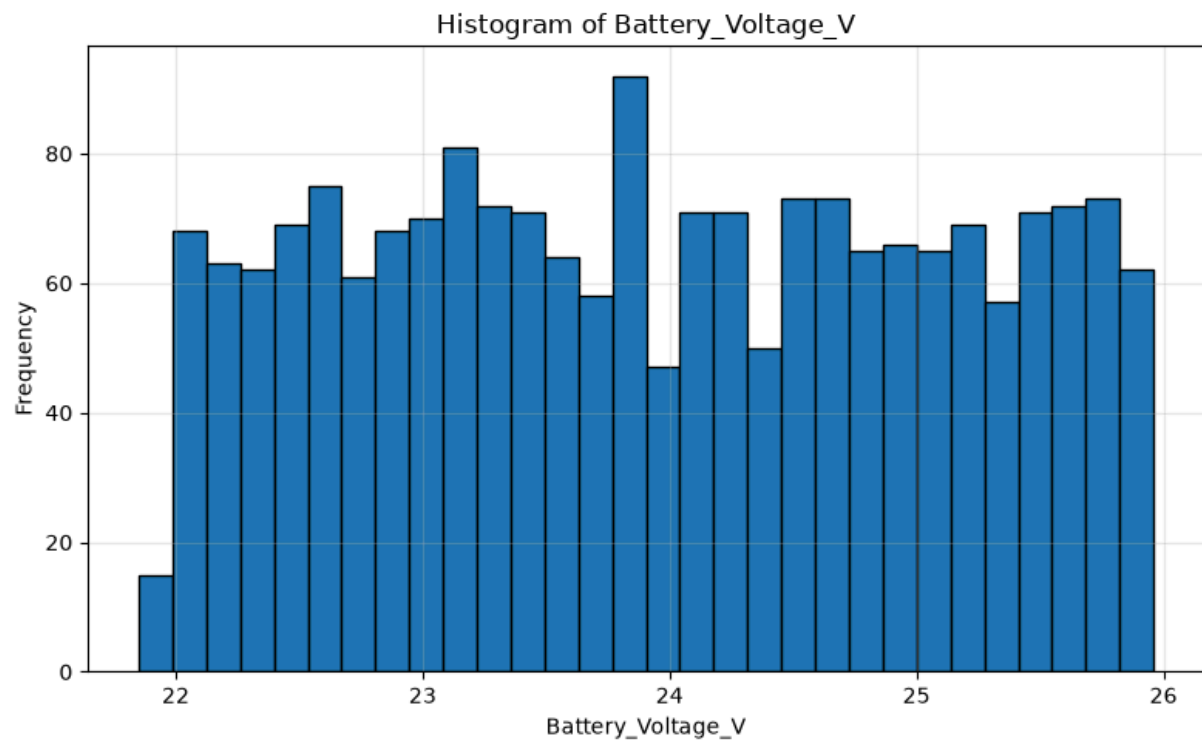
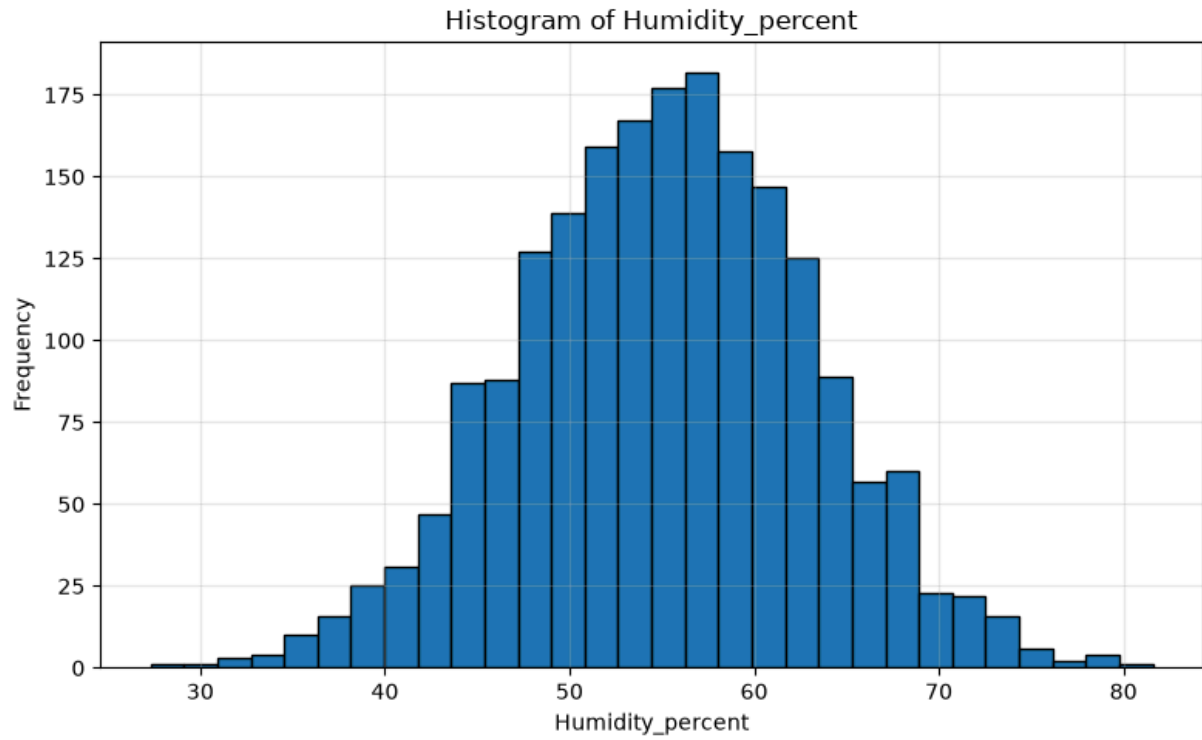
1. What is a probability distribution?
2. What is a normal distribution?
3. What is skewness?
4. What is kurtosis?
5. Why are histograms useful?
6. What is the difference between a population and a sample?
7. How can distributions help in detecting abnormal sensor readings?

Course Outcomes: (refer exp list)

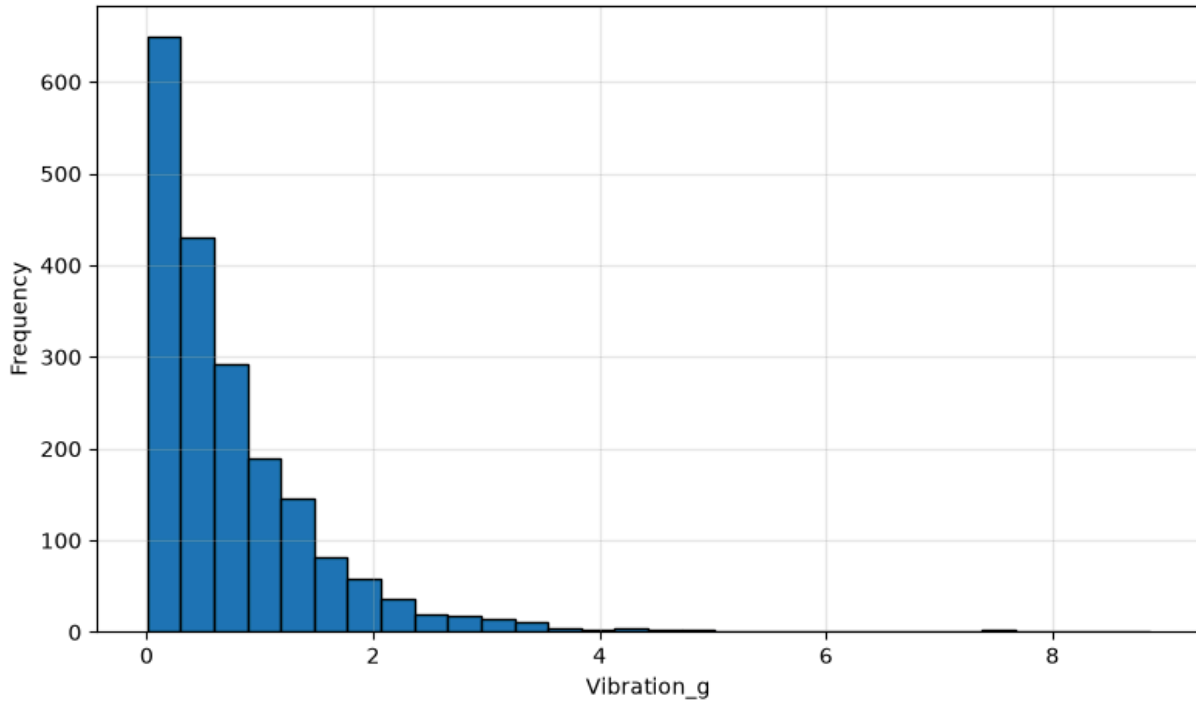
- **CO1:** Understand fundamental statistical concepts.
- **CO2:** Analyze statistical characteristics of datasets.
- **CO3:** Interpret probability distributions.
- **CO5:** Apply statistical knowledge to robotics data.

- Histogram Plot

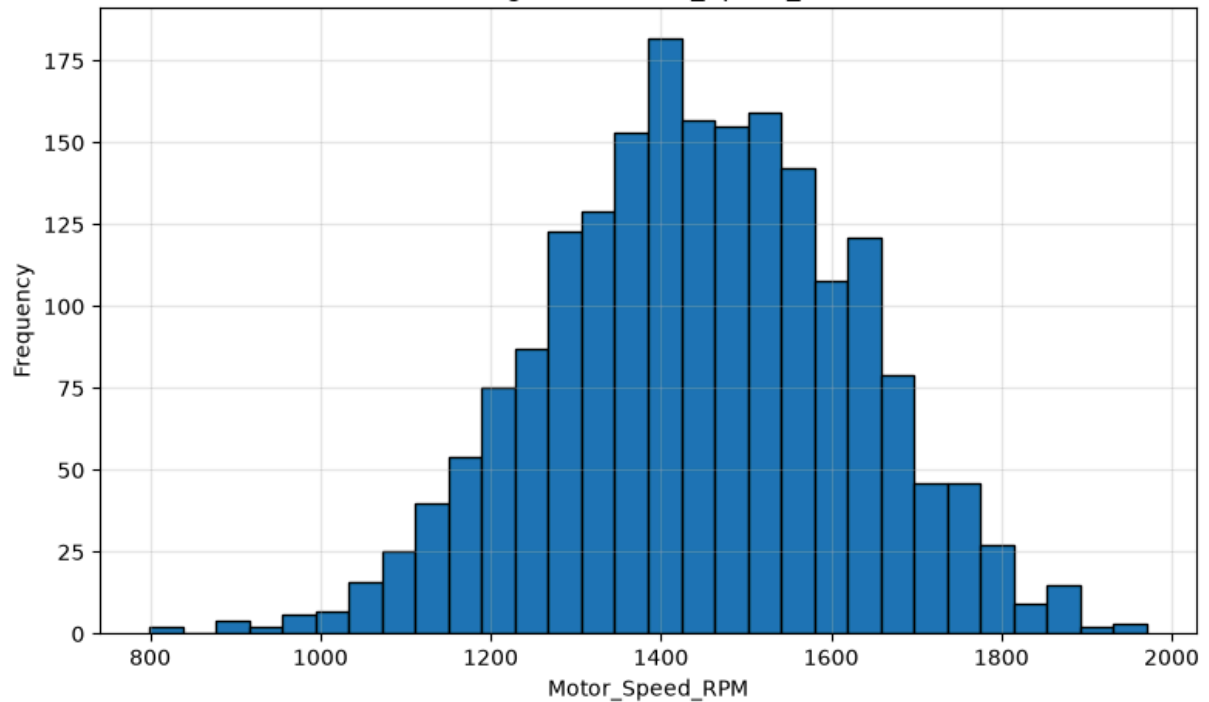




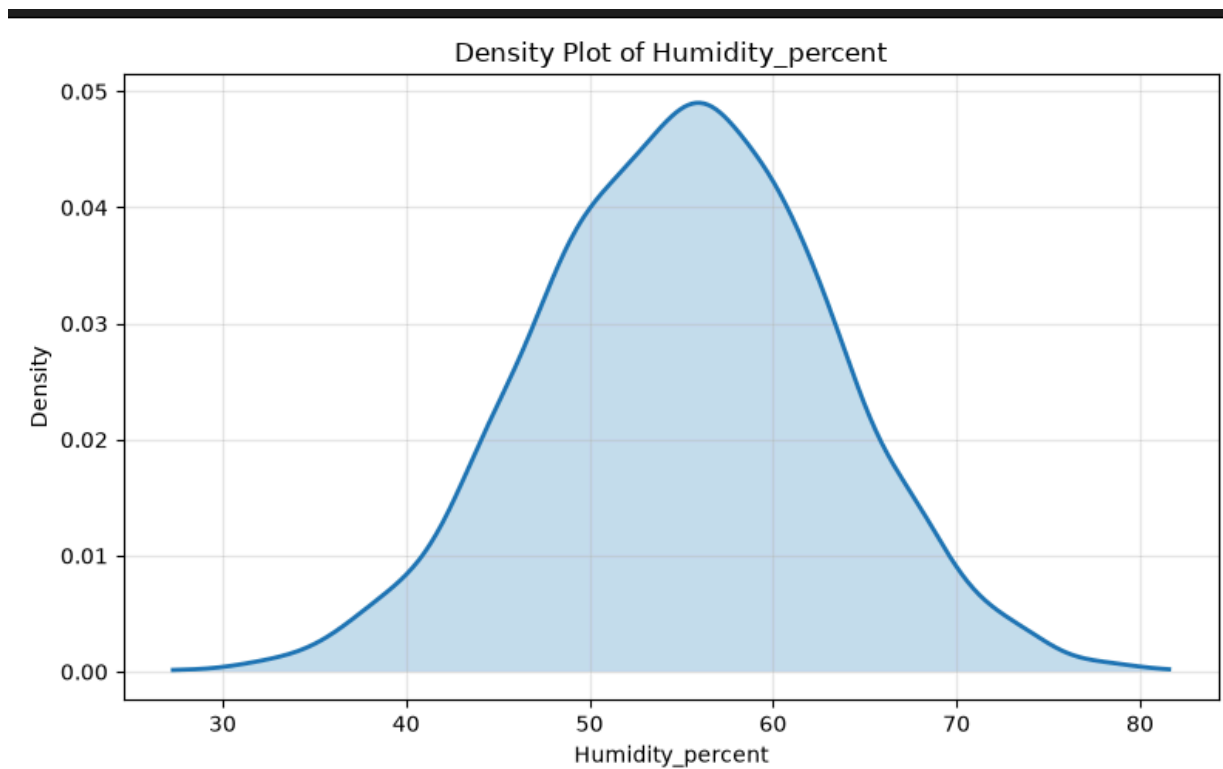
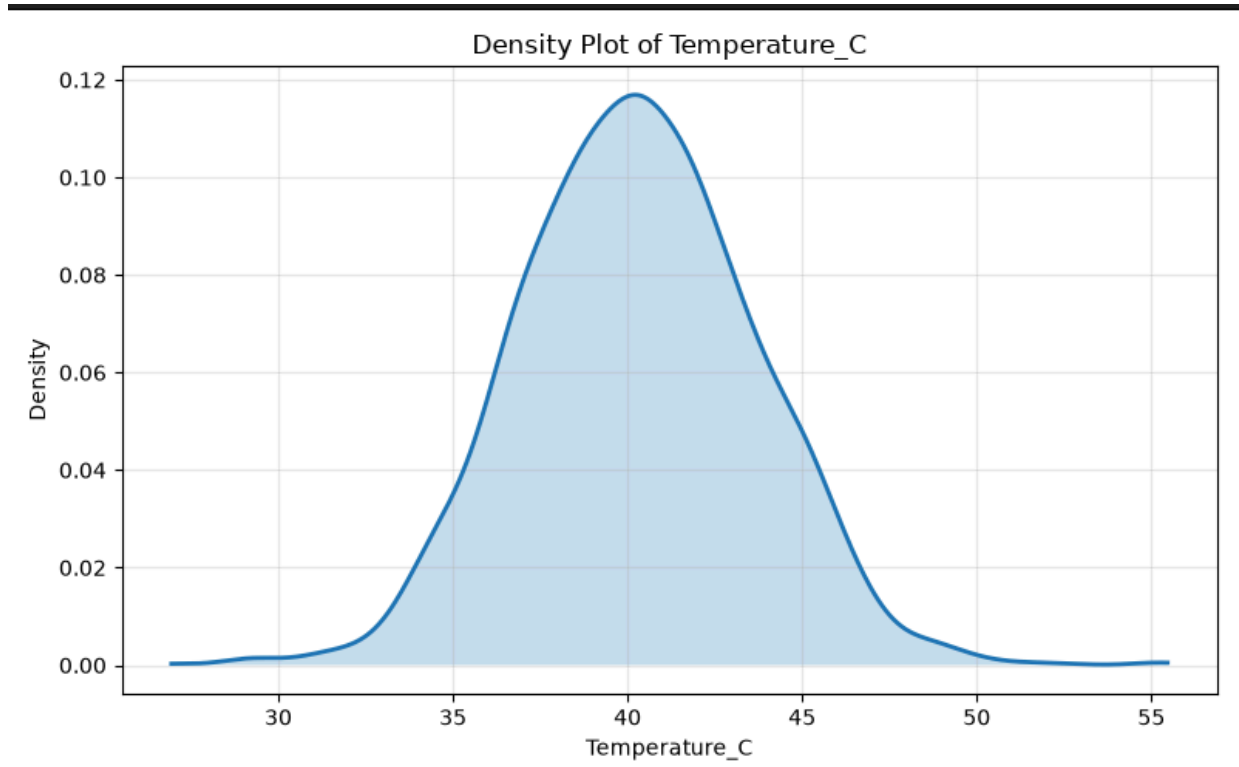
Histogram of Vibration_g

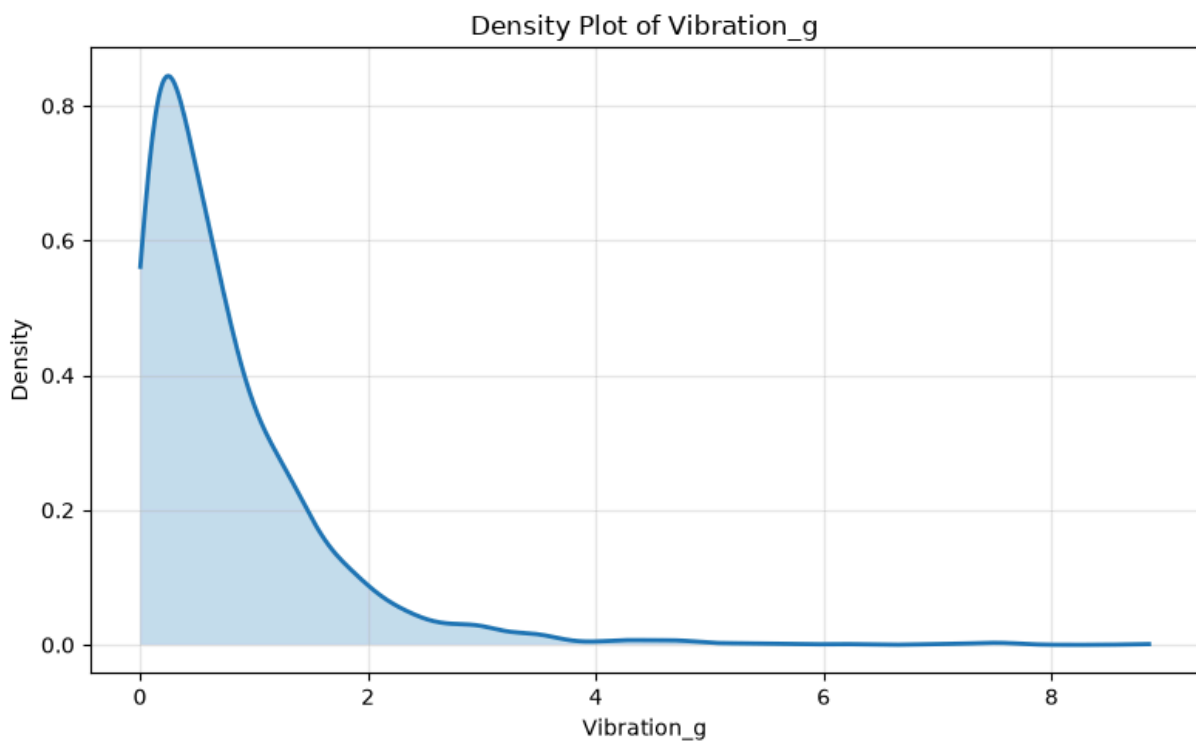
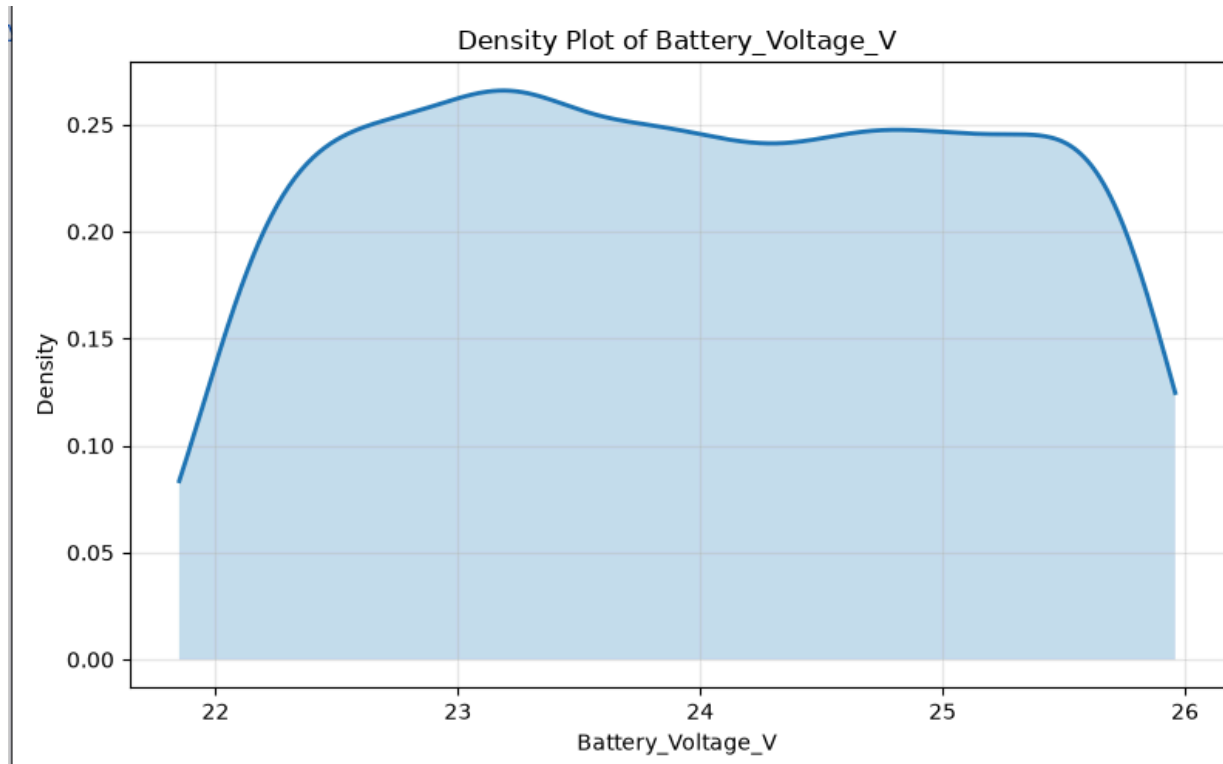


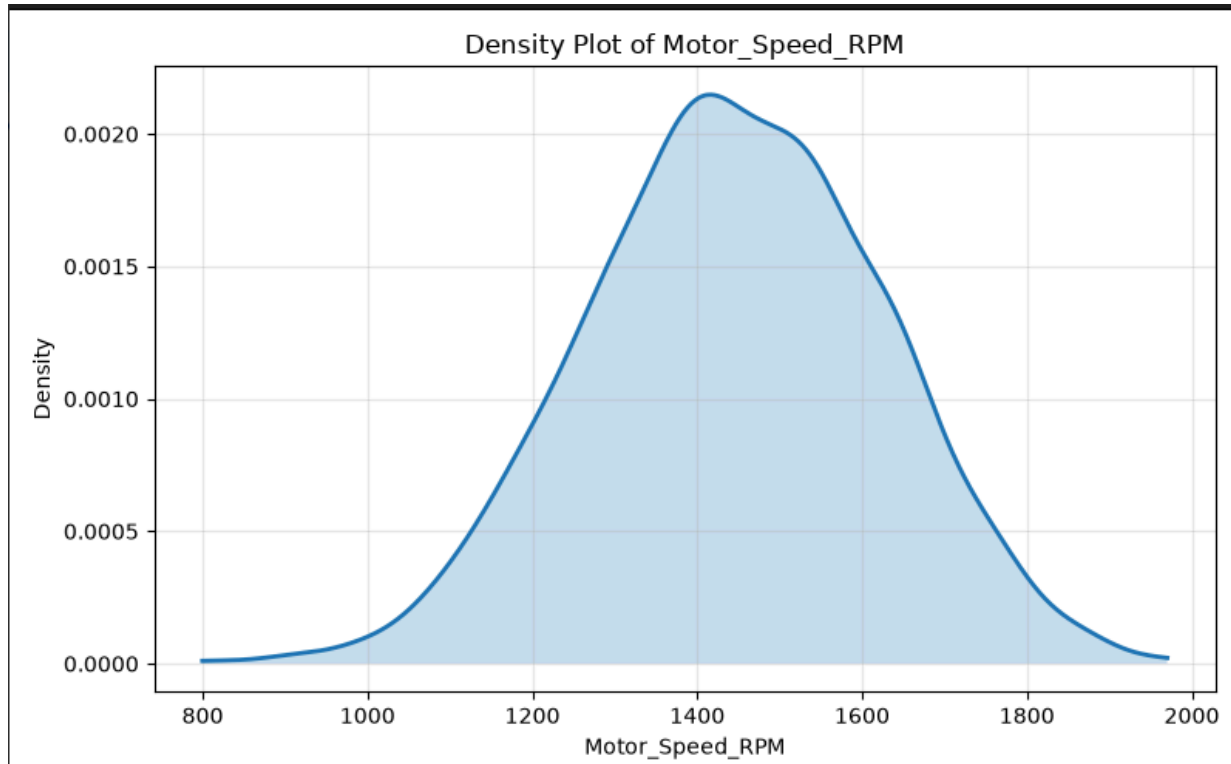
Histogram of Motor_Speed_RPM



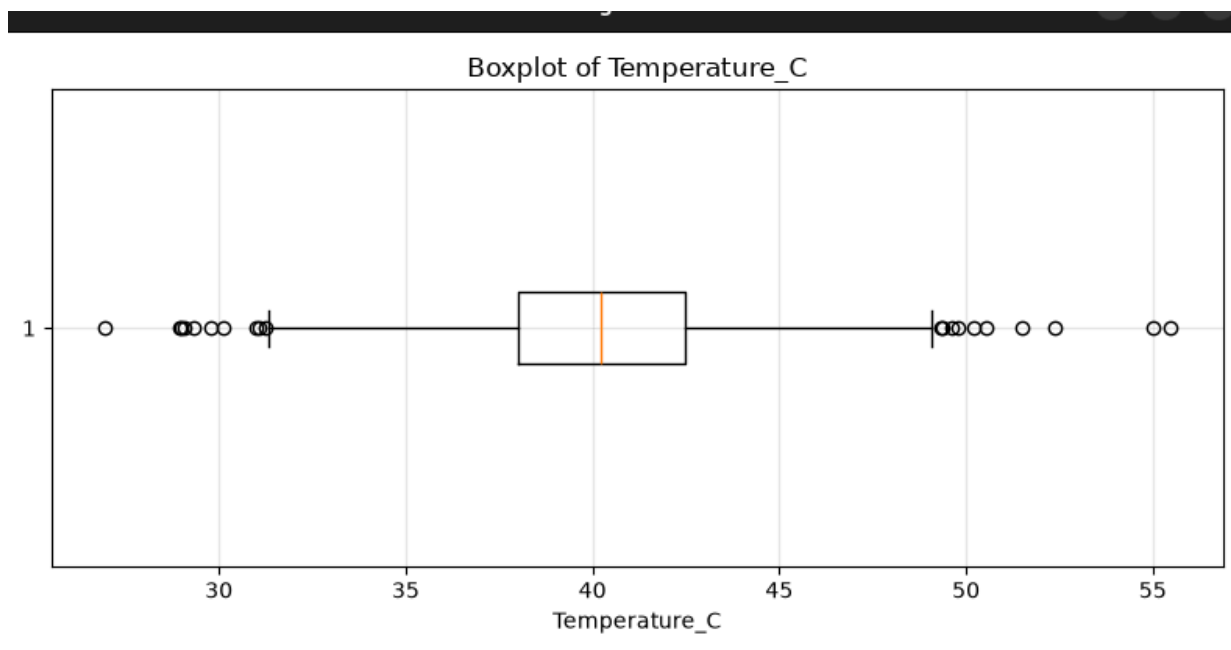
- Density Plot



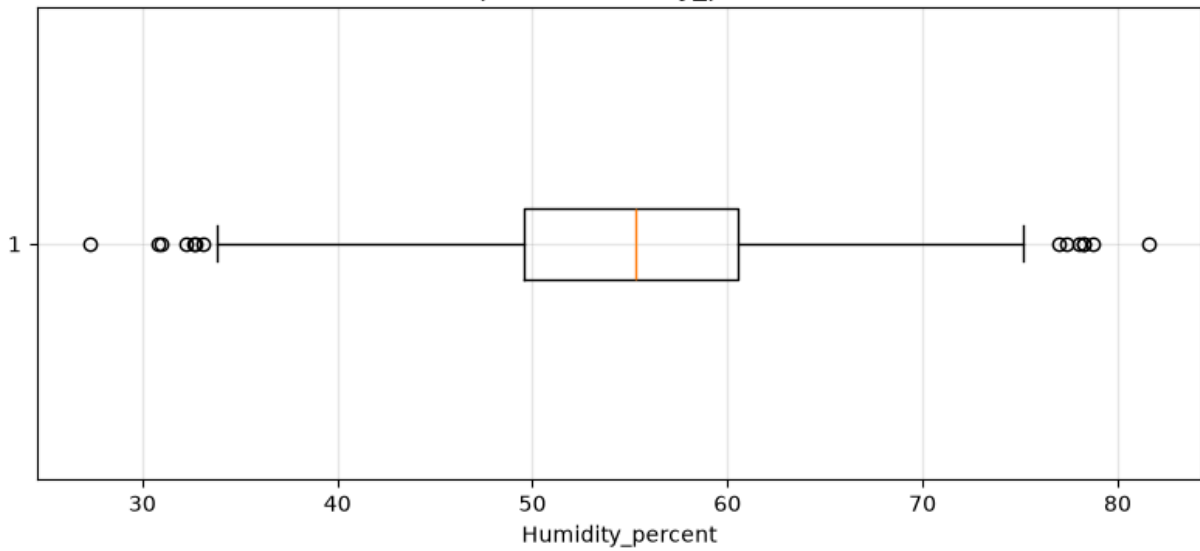




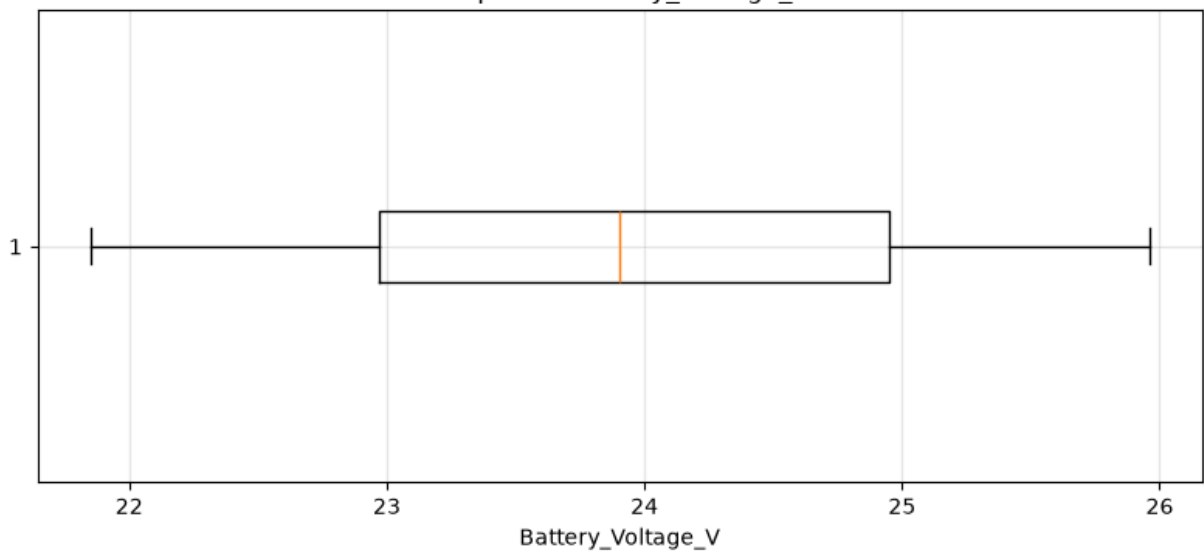
- Boxplot



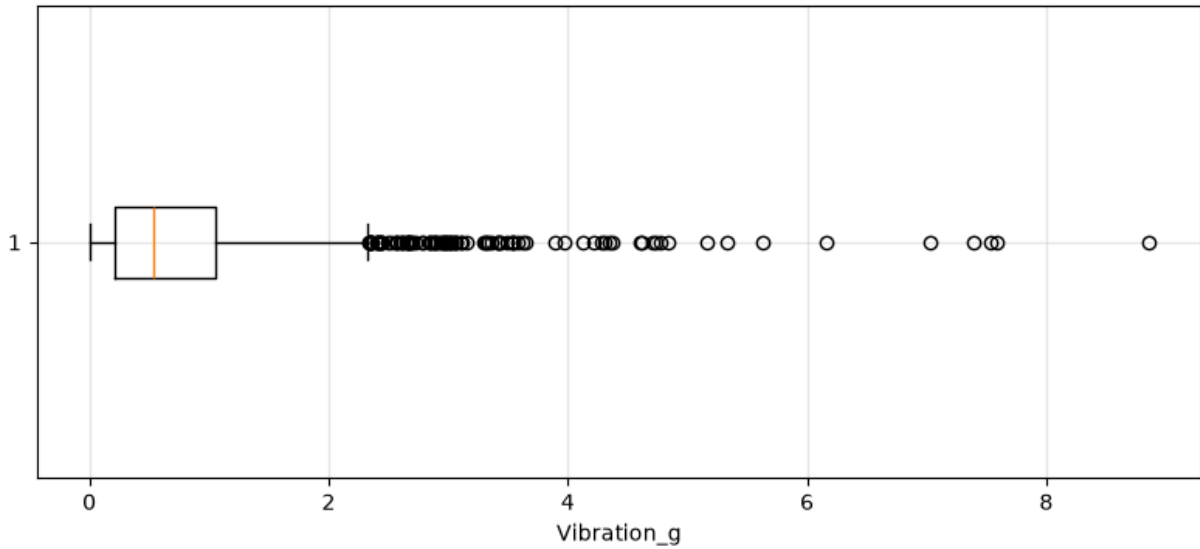
Boxplot of Humidity_percent



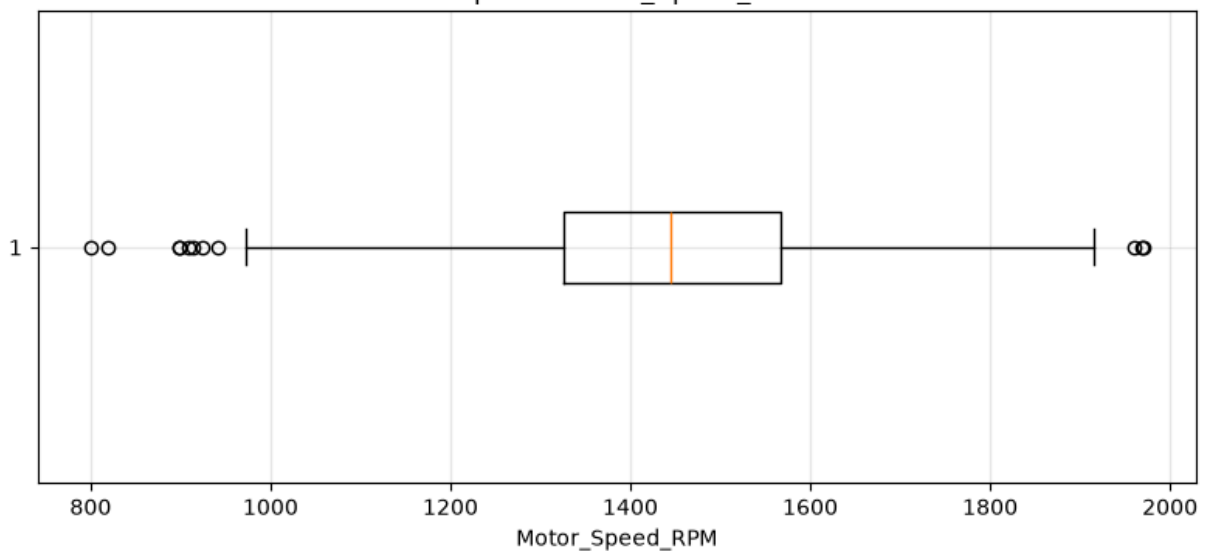
Boxplot of Battery_Voltage_V



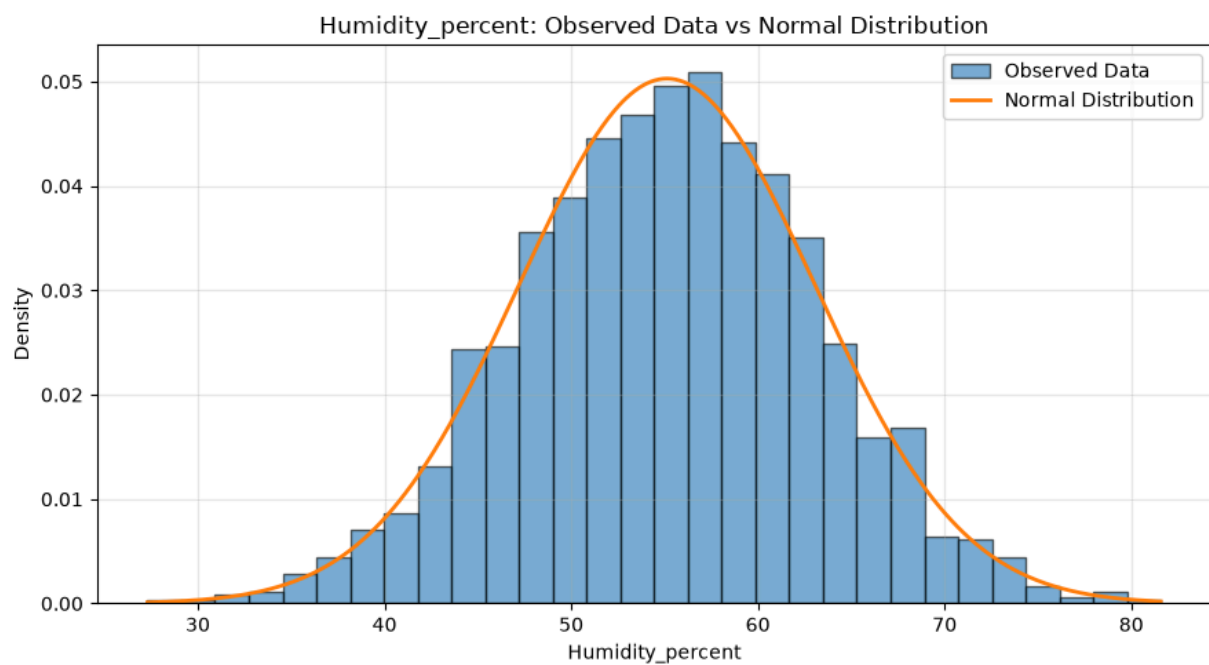
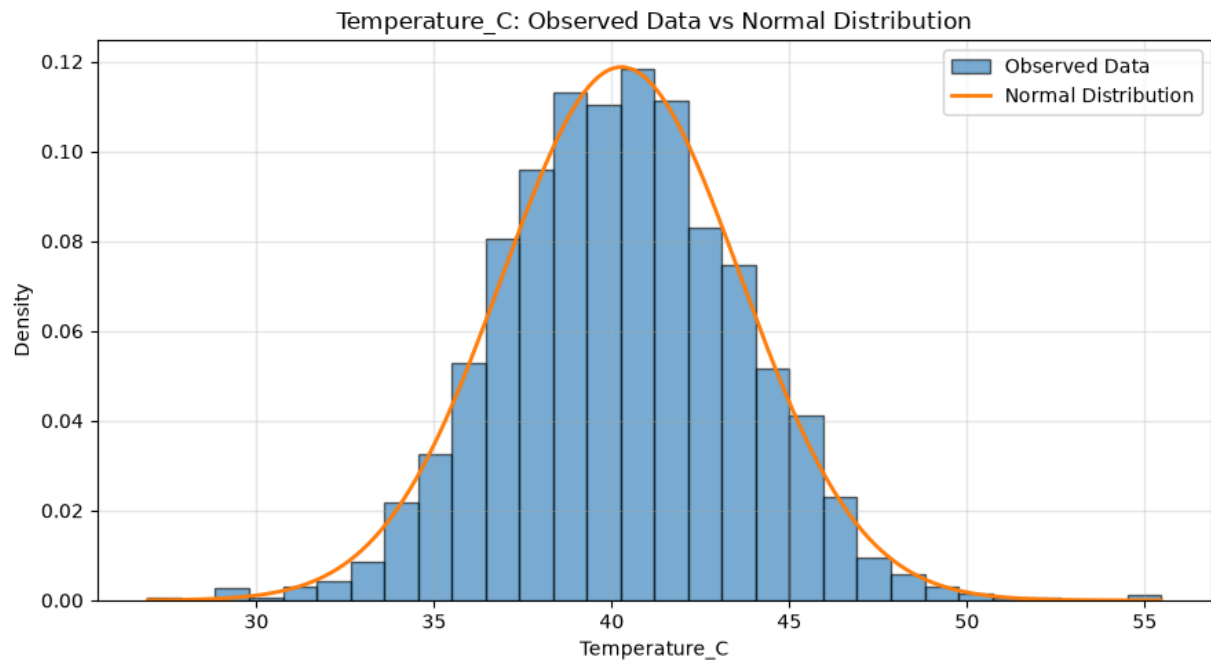
Boxplot of Vibration_g

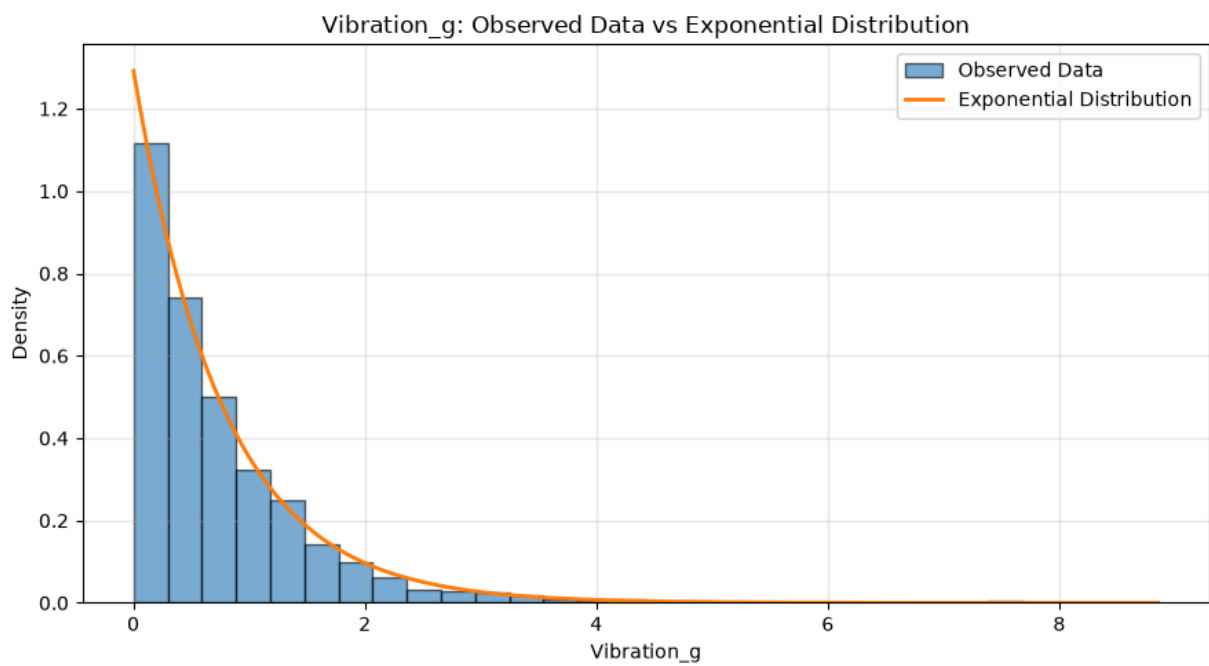
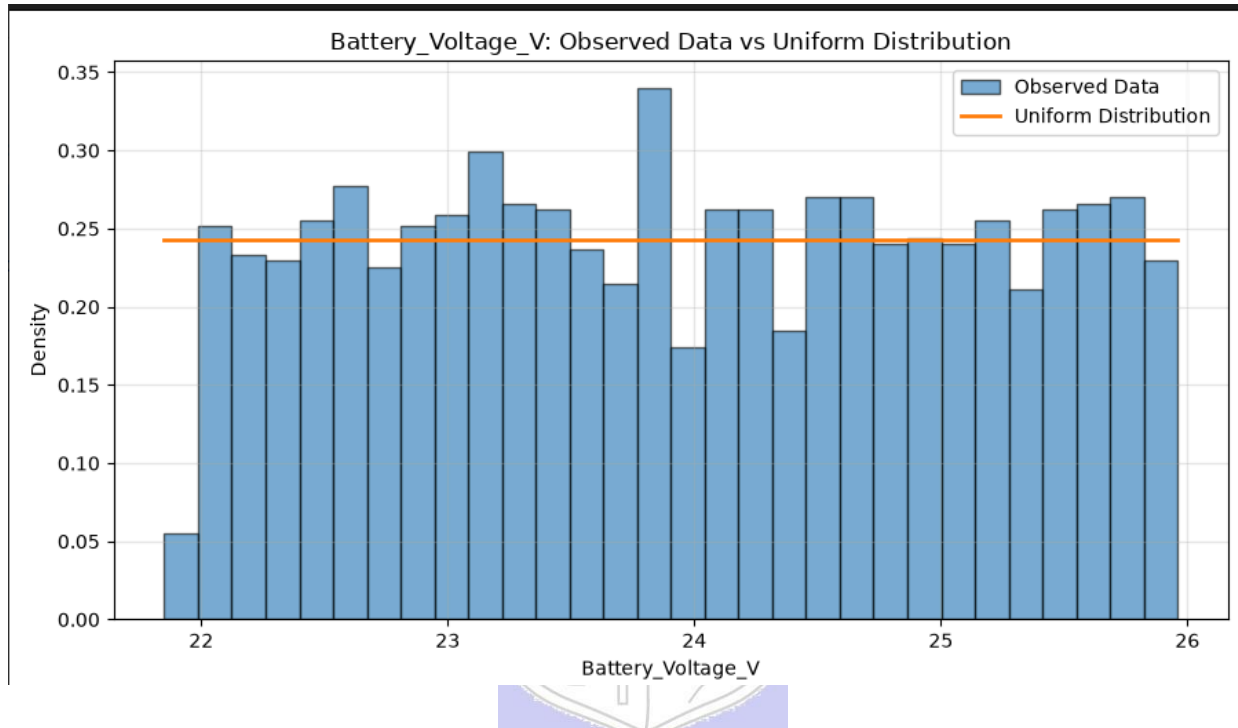


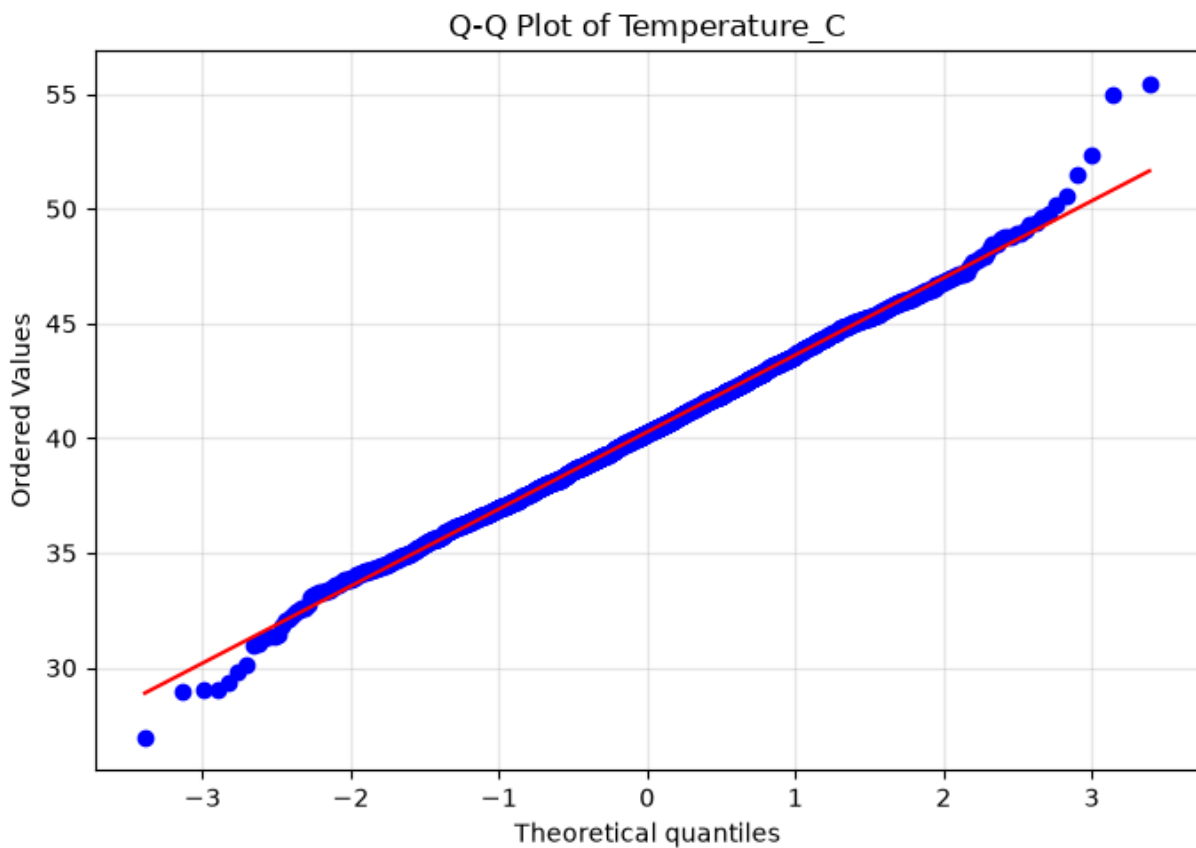
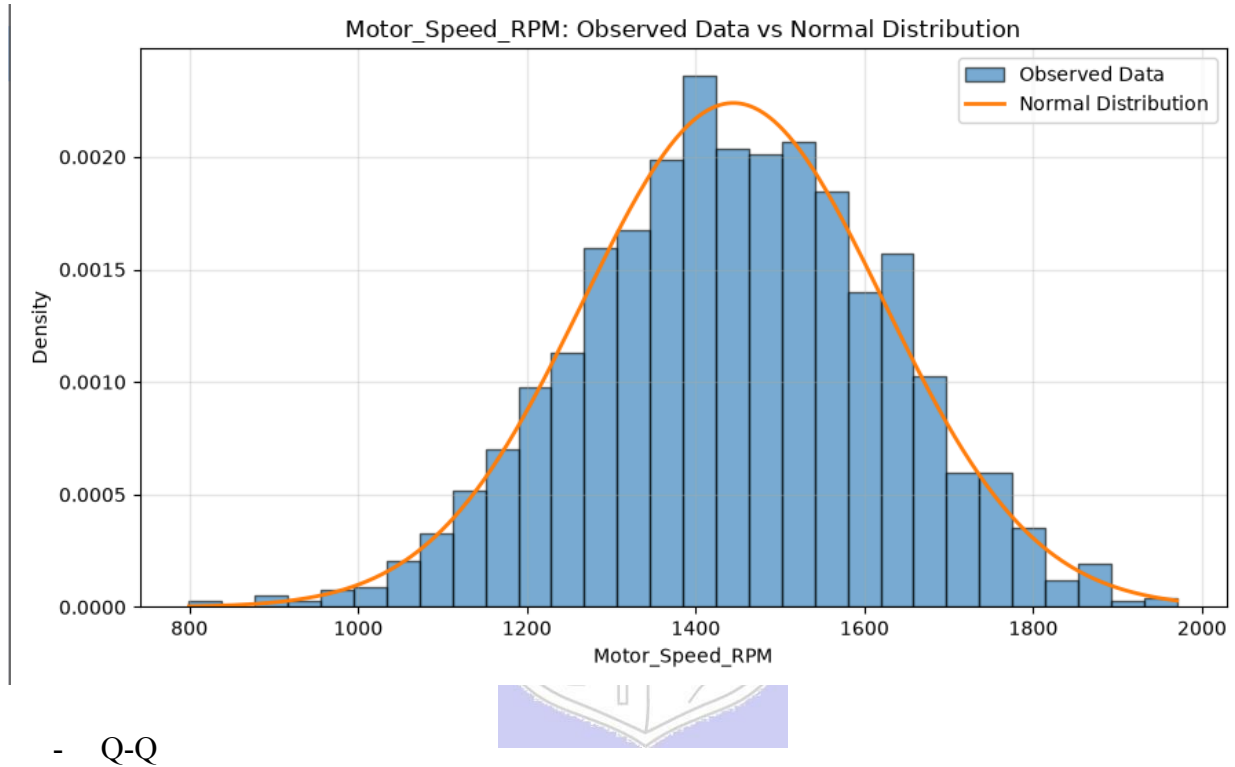
Boxplot of Motor_Speed_RPM



- Observed vs Normal







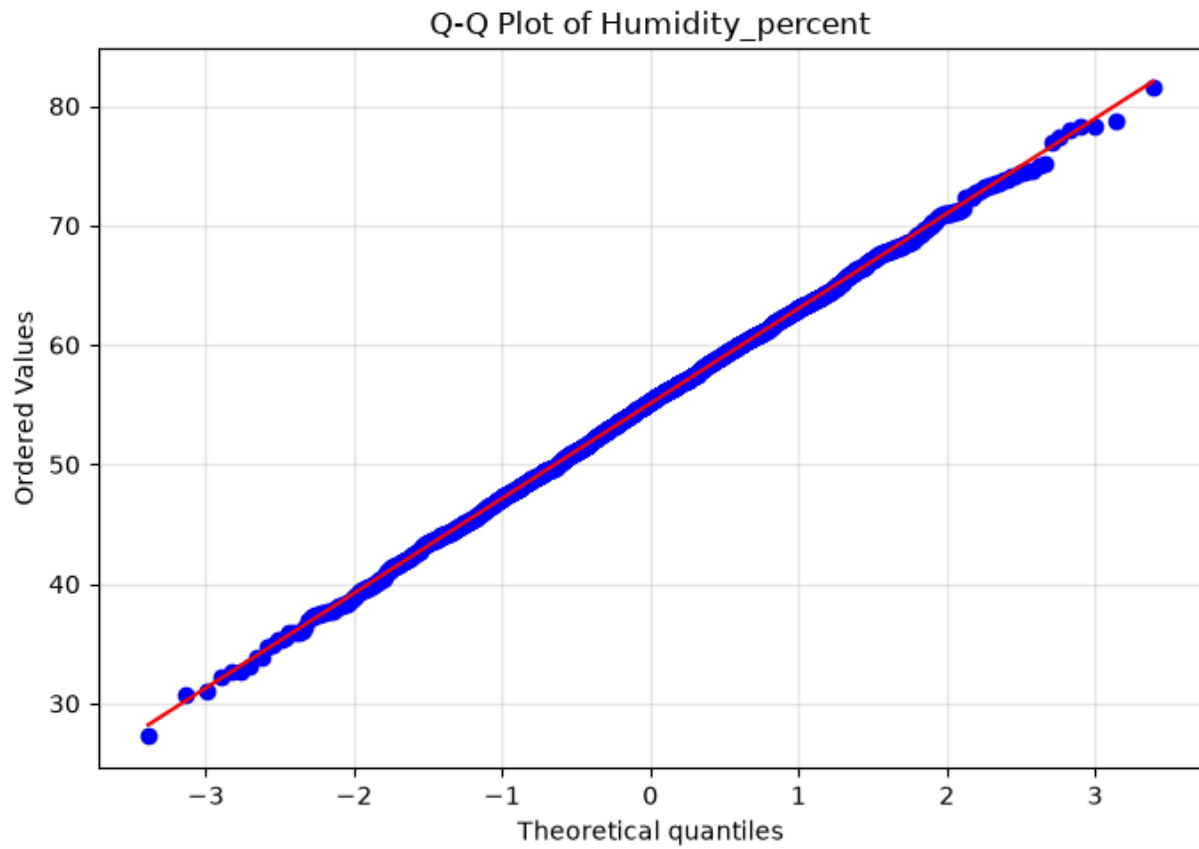


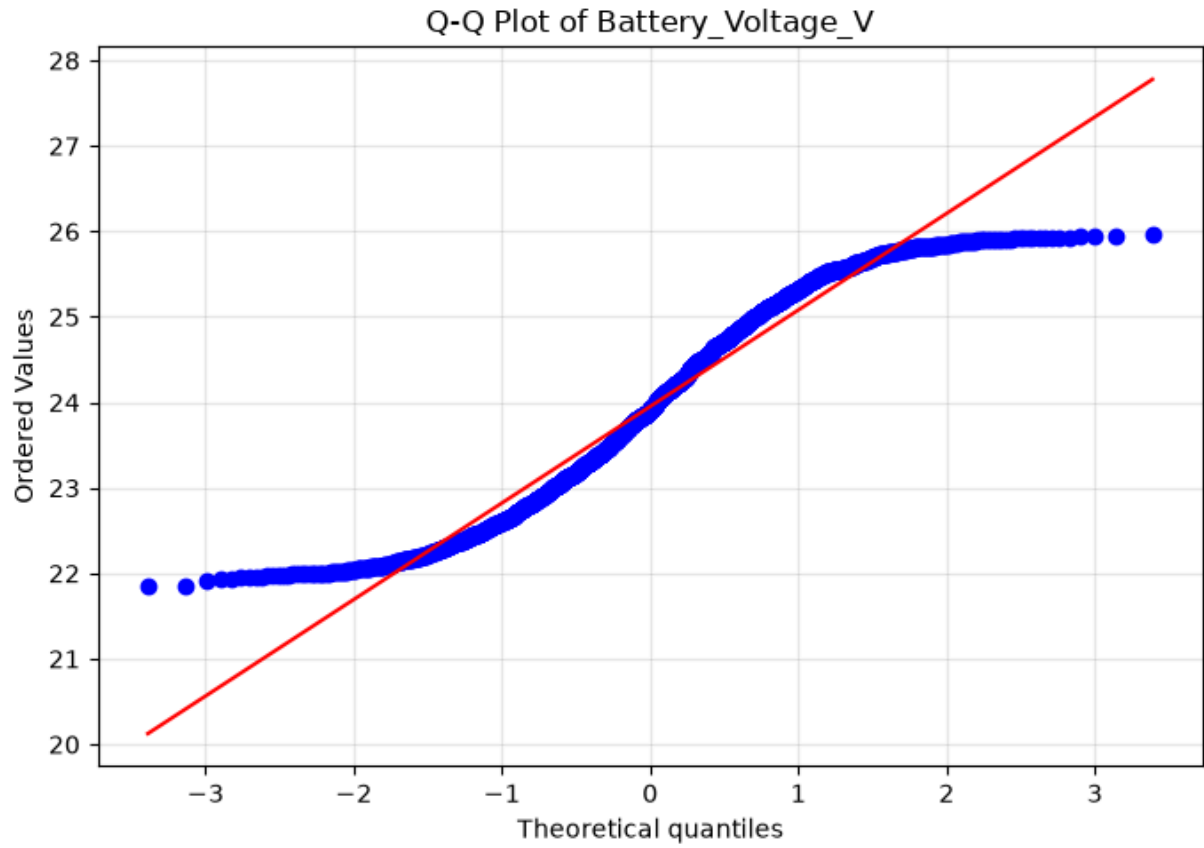
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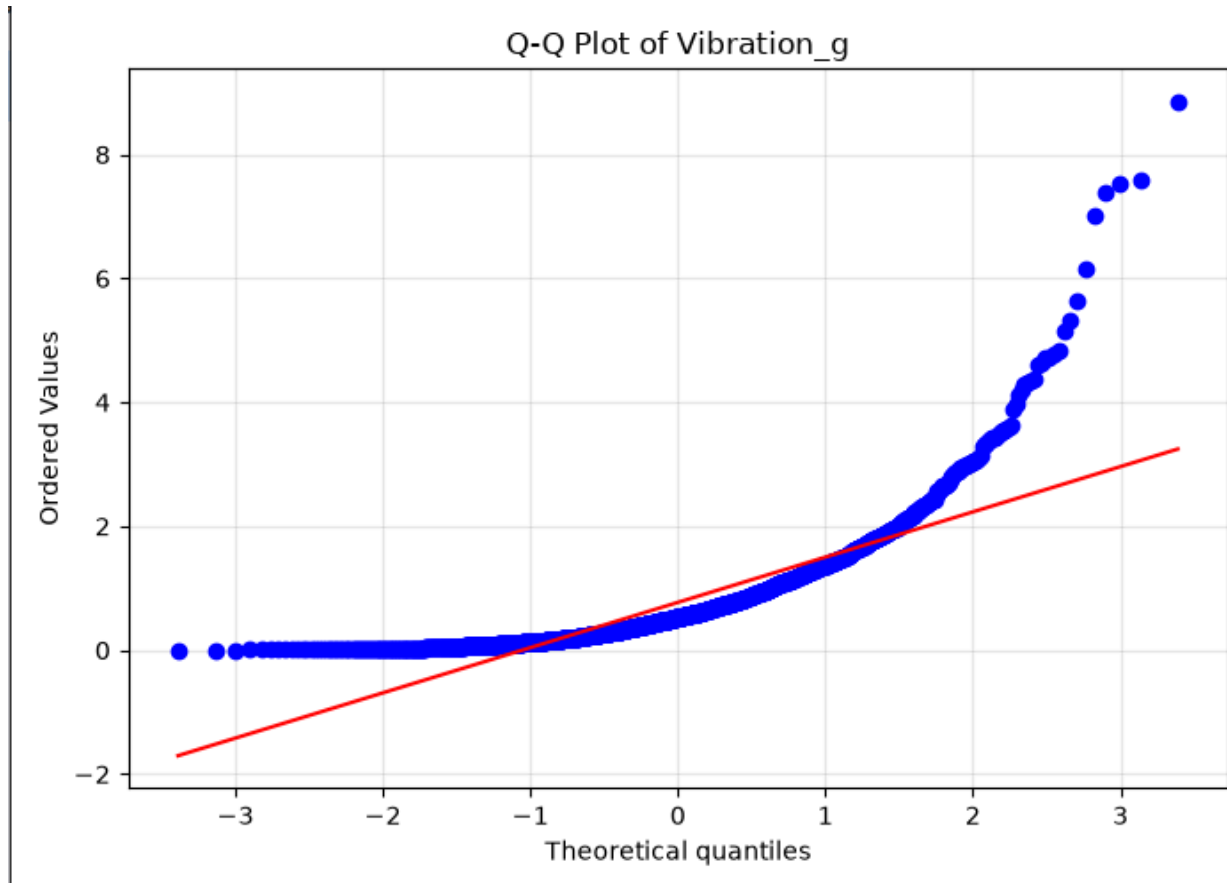
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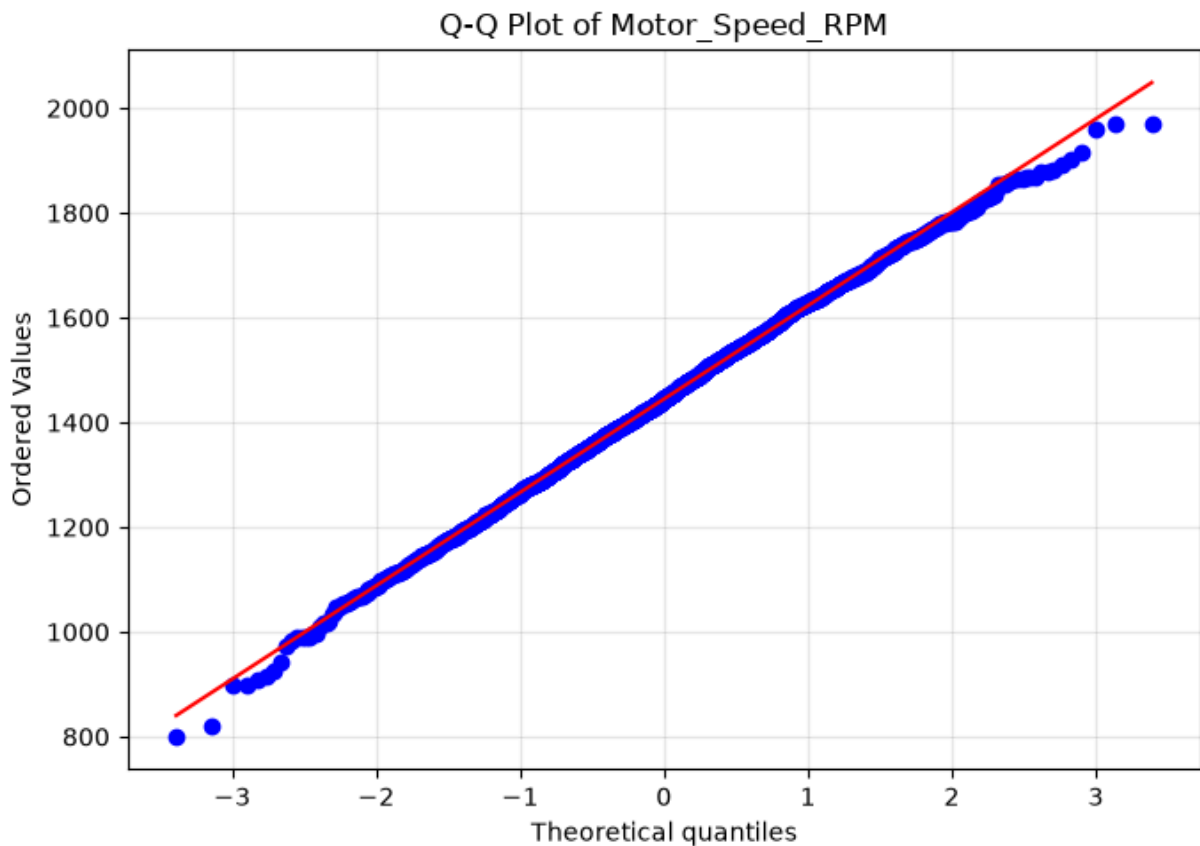


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Conclusion:

The statistical analysis showed that Temperature, Humidity and Motor Speed were approximately normally distributed, as indicated by their near-zero skewness, near-zero kurtosis and good agreement with the Normal distribution. Battery Voltage was better represented by a Uniform distribution, while Vibration followed an Exponential distribution and showed significant positive skewness and high kurtosis. Thus, the five robotic sensor variables do not all follow a normal distribution, demonstrating the importance of identifying the actual statistical distribution before applying statistical models to robotic sensor data.

Grade: AA / AB / BB / BC / CC / CD / DD**Signature of faculty in-charge with date**

References:



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- Douglas C. Montgomery and George C. Runger, *Applied Statistics and Probability for Engineers*.
- Jake VanderPlas, *Python Data Science Handbook*.
- SciPy documentation.
- Matplotlib documentation.

